

CS215 Assignment-1

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1. A Pretty “Normal” Mixture

Task A:

Solution:

$$\begin{aligned} F_A(c) &= P(A < c) \\ &= \sum_{i=1}^K P(A < c \wedge A = X_i) && \text{(By law of total probability)} \\ &= \sum_{i=1}^K P(A < c \mid A = X_i) \cdot P(A = X_i) \\ &= \sum_{i=1}^K P(X_i < c) \cdot P(A = X_i) \\ &= \sum_{i=1}^K p_i F_{X_i}(c) \end{aligned}$$

(Substituting the CDF $F_{X_i}(c)$ for each Gaussian component)

To find the probability density function (PDF) $f_A(c)$, we differentiate $F_A(c)$ with respect to c .

$$\begin{aligned} f_A(c) &= \frac{d}{dc} F_A(c) \\ &= \sum_{i=1}^K p_i \cdot \frac{d}{dc} F_{X_i}(c) \\ &= \sum_{i=1}^K p_i \cdot f_{X_i}(c) && \left(\frac{d}{dc} F_{X_i}(c) = f_{X_i}(c) \right) \end{aligned}$$

The PDF of the GMM variable X is f_X , given by $f_X(x) = \sum_{i=1}^K p_i \cdot f_{X_i}(x)$, which is exactly same as the one I got from my algorithm. Hence, $\forall u \in R, f_A(u) = f_X(u)$.

Task B:

Solution: Part 3:

$$\begin{aligned}M_X(t) &= E(e^{tX}) \\&= \int_{-\infty}^{\infty} e^{tx} f_X(x) \\&= \int_{-\infty}^{\infty} e^{tx} \cdot \left(\sum_{i=1}^K p_i \cdot f_{X_i}(x) \right) \\&= \sum_{i=1}^K \left(p_i \int_{-\infty}^{\infty} e^{tx} \cdot f_{X_i}(x) \right) \\&= \sum_{i=1}^K p_i E(e^{tX_i}) \\&= \sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}}\end{aligned}$$

We can now find the expected value $E[X]$ and variance $Var[X]$ using the MGF.

Part 1:

Differentiating the MGF with respect to t :

$$\begin{aligned}M'_X(t) &= \frac{d}{dt} \left(\sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \right) \\&= \sum_{i=1}^K p_i \cdot \frac{d}{dt} \left(e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \right) \\&= \sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \cdot (\mu_i + \sigma_i^2 t)\end{aligned}$$

Now, $E[X] = M'_X(0)$

$$\begin{aligned}E[X] &= M'_X(0) = \sum_{i=1}^K p_i \cdot e^0 \cdot \mu_i \\&= \sum_{i=1}^K p_i \cdot \mu_i\end{aligned}$$

Part 2:

$$\begin{aligned} M_X''(t) &= \sum_{i=1}^K p_i \cdot \frac{d^2}{dt^2} \left(e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \right) \\ &= \sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \cdot (\mu_i + \sigma_i^2 t)^2 + \sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \cdot \sigma_i^2 \end{aligned}$$

Now, the second moment $E[X^2]$ is:

$$E[X^2] = M_X''(0) = \sum_{i=1}^K p_i \cdot (\mu_i^2 + \sigma_i^2)$$

Now, the variance $\text{Var}(X)$ is:

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ \text{Var}(X) &= \sum_{i=1}^K p_i \cdot (\mu_i^2 + \sigma_i^2) - \left(\sum_{i=1}^K p_i \cdot \mu_i \right)^2 \end{aligned}$$

Task C:

Solution:

ATQ, $Z = \sum_{i=1}^K p_i X_i$.

Part 4:

$$\begin{aligned} M_Z(t) &= E[e^{tZ}] \\ &= E[e^{t \sum_{i=1}^K p_i X_i}] \\ &= E\left[\prod_{i=1}^K e^{tp_i X_i}\right] \\ &= \prod_{i=1}^K E[e^{tp_i X_i}] && \text{(As variables are independent)} \\ &= \prod_{i=1}^K e^{\mu_i p_i t + \frac{\sigma_i^2 p_i^2 t^2}{2}} \\ &= e^{\sum_{i=1}^K \mu_i p_i t + \frac{\sigma_i^2 p_i^2 t^2}{2}} \\ &= e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \end{aligned}$$

Part 1:

Differentiating $M_Z(t)$ with respect to t :

$$\begin{aligned} M'_Z(t) &= \frac{d}{dt} \left(e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \right) \\ &= e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \cdot \left(\sum_{i=1}^K \mu_i p_i + t \sum_{i=1}^K \sigma_i^2 p_i^2 \right) \end{aligned}$$

$$\begin{aligned} E[Z] &= M'_Z(0) = e^0 \cdot \left(\sum_{i=1}^K \mu_i p_i \right) \\ &= \sum_{i=1}^K \mu_i p_i \end{aligned}$$

Part 2:

$$\begin{aligned} M''_Z(t) &= \frac{d}{dt} \left(e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \cdot \left(\sum_{i=1}^K \mu_i p_i + t \sum_{i=1}^K \sigma_i^2 p_i^2 \right) \right) \\ &= e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \cdot \left(\left(\sum_{i=1}^K \mu_i p_i + t \sum_{i=1}^K \sigma_i^2 p_i^2 \right)^2 + \sum_{i=1}^K \sigma_i^2 p_i^2 \right) \end{aligned}$$

Evaluating at 0,

$$E[Z^2] = M''_Z(0) = e^0 \cdot \left(\left(\sum_{i=1}^K \mu_i p_i \right)^2 + \sum_{i=1}^K \sigma_i^2 p_i^2 \right) = \left(\sum_{i=1}^K \mu_i p_i \right)^2 + \sum_{i=1}^K \sigma_i^2 p_i^2$$

Now, the variance is:

$$\begin{aligned} \text{Var}(Z) &= E[Z^2] - (E[Z])^2 \\ &= \left(\sum_{i=1}^K \mu_i p_i \right)^2 + \sum_{i=1}^K \sigma_i^2 p_i^2 - \left(\sum_{i=1}^K \mu_i p_i \right)^2 \\ &= \sum_{i=1}^K \sigma_i^2 p_i^2 \end{aligned}$$

Part 3: PDF of Z

We observe that the MGF of Z matches the MGF of the gaussian $\mathcal{N}(\mu, \sigma^2)$ with $\mu = \sum_{i=1}^K \mu_i p_i$ and $\sigma^2 = \sum_{i=1}^K \sigma_i^2 p_i^2$.

$$Z \sim \mathcal{N} \left(\sum_{i=1}^K \mu_i p_i, \sum_{i=1}^K \sigma_i^2 p_i^2 \right)$$

Thus, the PDF of Z is:

$$f_Z(u) = \frac{1}{\sqrt{2\pi \sum_{i=1}^K \sigma_i^2 p_i^2}} \exp \left(-\frac{(u - \sum_{i=1}^K \mu_i p_i)^2}{2 \sum_{i=1}^K \sigma_i^2 p_i^2} \right)$$

Part 5:

No, X and Z have very different behaviors, and don't even have the same variances, much less the PDFs.

Part 6:

As concluded in part 3, Z is a gaussian.

Task D:

Solution: Let X be a finite discrete random variable taking values $x_1, x_2, \dots, x_n$ with respective probabilities $P(X = x_i) = p_i$, for $i = 1, 2, \dots, n$.

The moment generating function (MGF) of X , denoted by $\phi_X(t)$, is given by:

$$\phi_X(t) = E[e^{tX}] = \sum_{i=1}^n e^{tx_i} P(X = x_i) = \sum_{i=1}^n e^{tx_i} p_i$$

Consider the MGF $\phi_X(t)$. Since the MGF is a power series in t (the sum of exponentials of the discrete values x_i), it can be expanded as a Taylor series around $t = 0$:

$$\phi_X(t) = \sum_{k=0}^{\infty} \frac{E[X^k]}{k!} t^k$$

The coefficients of this series are the moments of the distribution, which are directly related to the probabilities $p_1, p_2, \dots, p_n$. If we know the MGF for all t in some interval, we can recover all the moments of X .

Since X takes a finite number of discrete values, the MGF is a finite sum of exponentials of t . The exponential function is one-to-one, so the MGF uniquely determines the probabilities $p_1, p_2, \dots, p_n$. Therefore, the MGF uniquely determines the distribution of X .

Step 3: Determining the PDF from the MGF

Given the MGF $\phi_X(t) = \sum_{i=1}^n e^{tx_i} p_i$, we can determine the probability mass function (PMF) by solving a system of linear equations. For distinct values $x_1, x_2, \dots, x_n$, the MGF can be written as:

$$\phi_X(t) = p_1 e^{tx_1} + p_2 e^{tx_2} + \dots + p_n e^{tx_n}$$

If we know $\phi_X(t)$ for enough values of t , we can solve for the probabilities $p_1, p_2, \dots, p_n$ by inverting this system. Therefore, the MGF uniquely determines the PMF.

Conclusion for Finite Discrete Case

Thus, for a finite discrete random variable, the MGF uniquely determines the distribution of X . Since the MGF and PDF (or PMF) uniquely determine each other, knowing the MGF is equivalent to knowing the distribution.

Conclusion about X and Z

Since Z is a weighted sum of independent Gaussian random variables and has the same MGF as X , they share the same distribution (by the theorem). This means that Z must also follow a Gaussian distribution with the mean and variance derived earlier.

Logical Explanation

The reason X and Z have the same distribution is that the MGF uniquely characterizes the distribution for both discrete and continuous variables. In this case, since Z is a sum of independent Gaussian variables, and Gaussian distributions are closed under linear combinations, Z follows a Gaussian distribution. The MGF, as a summary of all moments, ensures that Z has the same distribution as the sum of its components.